

EMERGENT DEPTHWISE ACTIVATION STRUCTURE IN TRANSFORMER LANGUAGE MODELS

THE HAU CURVE AND ITS EARLY-TRAINING ATTRACTOR PATTERN

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ABSTRACT

Large Language Models (LLMs) not only differ in performance, but also in how they distribute computation across depth. In this work, we identify a remarkably consistent three-phase activation pattern that emerges during training and recurs across diverse decoder-only transformers. Analyzing layer-wise ℓ_1 -activations on paired abstract and literal prompts for 60 models, we find that sufficiently large architectures exhibit a conserved depth profile characterized by an early-layer extremum, a suppressed mid-layer plateau, and a late-layer surge—a structural motif we term the Hau Curve. This pattern appears as the dominant mode of variation in a principal component analysis (PC1 accounting for roughly one-third of variance in dense-condition activations). Developmental analyses show that it arises early in pretraining ($\approx 7,000$ steps for a 12B model) and then stabilizes, suggesting a training-induced attractor in network dynamics. Notably, the effect is weak or inconsistent in architectures with fewer than ~ 20 –24 transformer blocks, and stabilizes in models with ≥ 30 –32 blocks, indicating a depth-dependent emergence threshold rather than a purely parametric one. Together, these findings position the Hau Curve—and its convergence behavior, the Hau Attractor—as fundamental structural properties of transformer LLMs, revealing intrinsic architectural dynamics not observable from model behavior alone.

1 Introduction

Large language models (LLMs) exhibit remarkable behavioral capabilities, yet most evaluations focus on their external outputs—benchmark scores, task accuracy, or alignment behavior. Such metrics, while useful, provide limited insight into a model’s internal computational structure. Small changes in prompt phrasing, sampling settings, or instruction tuning can shift behavioral performance, making it difficult to determine whether two models differ in their underlying mechanisms or merely in their surface-level responses.

In contrast, the mechanistic interpretability community has uncovered localized internal structures within transformers, such as induction heads, key–value memory patterns in MLP layers, and specialized attention circuits. These findings reveal that models contain reproducible internal components, but most such work focuses on isolated features or layers rather than on global architectural organization. At present, it remains unclear whether transformer models exhibit stable, model-specific patterns in how they allocate computation across depth.

To investigate this possibility, we adopt an internal, activation-based approach. Rather than evaluating models through their generated text, we measure how activation magnitude is distributed across layers when models process inputs that differ systematically in semantic abstraction. Drawing inspiration from cognitive theories of depth of processing, we probe each model using paired prompts that express the same idea in two ways: a semantically rich (“dense”) form and a literal (“flat”) form. The differences between these conditions serve as a controlled lens through which to examine how each network reallocates computation across its depth.

Initial exploratory experiments revealed that every model tested (excluding small control models) exhibited consistent depth shifts between abstract and literal conditions. More surprisingly, the *shape* of these layerwise activation differences was far from random. Each model expressed a distinctive, stable depthwise pattern when processing abstract inputs,

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and these patterns persisted across prompt variations and hardware platforms. Motivated by these observations, we constructed a fully standardized prompt set to isolate depth structure with minimal syntactic or length variation and then systematically evaluated models across a wide range of architectures and scales.

This analysis exposed a striking regularity: many large transformer LLMs share a highly stable, three-phase depthwise profile characterized by (1) an early-layer extremum (a sharp positive spike), (2) a suppressed mid-layer plateau, and (3) a late-layer surge. We term this shared structure the *Hau Curve*. Moreover, by examining pretraining checkpoints from two unrelated model families, we find that this curve emerges early in training and then stabilizes—a developmental trajectory we refer to as the *Hau Attractor*.

While earlier analyses suggested a parameter-scale threshold, the expanded cross-model evaluation demonstrates that architectural depth—rather than total parameter count—is the primary determinant of whether the three-phase structure fully emerges.

This work introduces a *structural psychometrics* perspective: characterizing models by their intrinsic activation dynamics rather than by their externally observable behavior. Our contributions are as follows:

1. We show that semantic abstraction reliably modulates depthwise activation in a consistent direction within each architecture.
2. We demonstrate that large decoder-only transformers express a highly reproducible depthwise activation signature—the *Hau Curve*—across diverse model families and training regimes.
3. Using principal component analysis (PCA), we identify this three-phase pattern as the dominant structural mode of variation across models.
4. Through checkpoint analysis, we show that the pattern emerges early in pretraining and remains stable, suggesting the presence of a training-induced depthwise attractor.
5. We outline how this structural perspective complements existing behavioral and mechanistic approaches, offering invariant internal metrics for comparing architectures and analyzing training dynamics.

2 Related Work

Prior work relevant to this study spans several research areas, including emergent behaviors in large-scale transformer training, behavioral evaluation, mechanistic interpretability, representational depth, and cognitive theories of abstraction. We briefly summarize key developments in each domain. The following section will discuss how the present work addresses these.

2.1 Emergent Phenomena in Large-Scale Transformer Training

A growing body of work has documented emergent behaviors that arise during large-scale transformer pretraining. Scaling laws show that model performance, loss gradients, and representational structure follow predictable trajectories as parameters, data, and compute increase [1]. More abrupt transitions have also been observed: models may undergo phase changes in their internal organization during training, such as the sudden appearance of algorithmic capabilities (“grokking”) after extended optimization [2]. Mechanistic studies have revealed similarly discrete formation events, such as the emergence of induction heads—attention circuits that appear reliably at certain training stages and scales [3]. These findings suggest that transformers develop characteristic internal structures as a function of scale and training progression.

2.2 Behavioral Evaluation

Large language models are typically evaluated through behavioral outputs such as benchmark scores, instruction-following quality, and few-shot performance [4, 5, 6, 7].

2.3 Mechanistic Interpretability and Internal Circuit Analysis

Mechanistic interpretability research seeks to identify the circuits and computations implemented by transformer models. Prior work has mapped attention head roles, key–value memory structures in MLP layers, and localized functional units [8, 9, 10, 11]. More recent studies investigate superposition, sparsity, and monosemanticity as organizing principles of feature representations [12].

2.4 Representational Depth, Layer Roles, and Probing Methods

Several studies have examined how linguistic features distribute across the layers of transformer architectures. Probing analyses have revealed that lower layers encode syntactic structures, while deeper layers capture semantic and contextual information [13, 14, 15, 16]. Additional work has explored redundancy, information flow, and layer specialization in pretrained models [17, 18]. Parallel research on contextual embeddings [19] and representational geometry [20, 21] has highlighted how internal representations evolve with depth. These investigations establish that transformer layers differ in function, but they do not typically measure how *processing effort reallocates* across layers when models are challenged with more semantically dense input.

Recent theoretical work has also proposed three-phase organizational principles for transformer computation. The Mix–Compress–Refine (MCR) framework links attention sinks and compression valleys to residual-stream dynamics, suggesting early mixing, mid-layer compression, and late-layer refinement [22]. Complementary spectral analyses of layer-wise Jacobians identify analogous stages—feature extraction, representation refinement, and output preparation—emerging from singular-value dynamics [23]. These approaches focus on token-level information flow or local linear behavior, whereas the Hau Curve captures a global, model-level activation geometry that recurs across architectures and scales. The convergence of these independent perspectives suggests that three-phase depth organization is a fundamental property of transformer training dynamics, manifesting at behavioral, spectral, and activation-structural levels.

2.5 Cognitive Foundations: Abstractness, Concreteness, and Depth of Processing

The distinction between abstract and concrete language has deep roots in cognitive psychology. Classic theories such as Levels of Processing (LOP) propose that semantic depth influences encoding strength [24]. Complementary frameworks such as Dual Coding Theory [25] and concreteness effects [26, 27] demonstrate that abstract and metaphorical language engages more distributed neural resources in humans.

2.6 Structural Metrics and Internal Model Characterization

Internal model characterization has primarily focused on probing hidden representations, analyzing embedding geometry, or identifying functional circuits within specific layers [20, 19, 17]. Although these approaches reveal important properties of transformer computations, they typically emphasize local phenomena—such as individual neurons, attention heads, or specific representational subspaces—rather than global activation structure across the full architecture.

A smaller set of studies uses activation magnitude or activation differences as analytic signals. Feed-forward layers have been shown to function as key–value memory systems with identifiable activation patterns [10]. Other work uses activation vectors to locate or modify semantic associations [28], or to compare representational similarity across layers using SVCCA or PWCCA [20]. These methods establish that activation geometry encodes meaningful structure, but they do not attempt to model depthwise allocation of activation or characterize architecture-scale invariants in activation patterns.

3 Positioning of the Present Work

The domains summarized above highlight substantial progress in understanding transformer models from behavioral, probing-based, and circuit-level perspectives. They also reveal several open gaps. Behavioral evaluations provide useful insights but are highly sensitive to prompt phrasing and sampling choices, making it difficult to infer stable internal properties from output alone. Probing methods characterize information present in hidden states but typically target individual layers or subspaces rather than global architectural structure. Mechanistic interpretability identifies specific components such as attention heads or MLP features, but focuses on localized mechanisms rather than model-wide activation behavior. Finally, research on emergent phenomena has documented discrete capability transitions during training, yet has not examined whether there exist global activation patterns that emerge across architectures.

The present work addresses these gaps by adopting an internal, activation-based perspective. Rather than evaluating a model solely through its generated outputs or isolated representational features, we analyze layer-wise activation magnitudes elicited by matched abstract and literal prompts. This approach enables us to study how transformer models allocate computational depth under controlled semantic variation, providing a view of processing that is less dependent on prompt formatting or sampling variability.

Our goal is to characterize architecture-scale activation structure: how activation magnitude is distributed across depth, how it changes in response to semantic complexity, and whether consistent patterns arise across model families and

training stages. This activation-centric perspective complements existing behavioral and mechanistic approaches and motivates the methodological framework introduced in the following section.

4 Methodology

4.1 Experiment Design

Our experimental design proceeded in four phases, each addressing a distinct methodological question and building on insights from earlier phases.

4.1.1 Phase 1 — Depth Processing Inquiry

The initial phase examined whether semantic abstractness influences depthwise activation patterns across transformer architectures. We evaluated 14 models ranging from $< 70\text{M}$ to $\sim 34\text{B}$ parameters, including two very "debugging controls" (GPT-2 and EleutherAI/Pythia-70M) used to validate the extraction pipeline. A heterogeneous stimulus set containing abstract ("dense") and literal ("flat") prompt pairs spanning diverse topics and linguistic forms. Prompts varied in metaphorical content, syntactic structure, and rhetorical presentation. Phase 1 functioned as an exploratory probe into whether semantic richness produces systematic shifts in internal depth allocation.

4.1.2 Phase 2 — Controlling Confounds with Declaratives Only (Δ_s)

Findings from Phase 1 motivated a more controlled stimulus design. Phase 2 employed a strictly matched set of metaphor–literal pairs. Because Phase 1 mixed declarative and interrogative prompts—introducing a secondary syntactic variable—Phase 2 restricted all prompts to declaratives only. The objective was to isolate semantic abstraction as the sole varying factor and establish a stable measurement environment for all subsequent experiments.

4.1.3 Phase 3 — Cross-Model Structural Characterization

Phase 3 scaled the analysis to a wide range of transformer architectures. For models already evaluated in Phase 2 (14 total), we reused their Δ_{layer} curves.

An additional 31 large models (7B–34B parameters) were evaluated using the same extraction pipeline to ensure consistency. All models in this phase were processed on the same GPU hardware, using the Phase 2 stimulus set for comparability.

A secondary objective of Phase 3 was to investigate developmental emergence. To this end, a lightweight second test suite processed intermediate pretraining checkpoints for two families: allenai/OLMo-2-1124-13B and EleutherAI/pythia-12b-deduped. Checkpoint evaluations likewise used the Phase 2 stimulus set to ensure comparability with the fully trained models.

In total, Phase 3 encompassed the 14 Phase 2 models plus 31 new large models (45 large models total), spanning multiple families (LLaMA, Qwen, Mistral, DeepSeek, OLMo). This phase aimed to determine:

- whether depthwise activation patterns vary across architectures, scales, or alignment regimes, and
- whether stable structural tendencies emerge during pretraining.

4.1.4 Phase 4 — Emergence Conditions in Small Models

Phase 4 investigated the conditions under which the depthwise activation structure (the Hau Curve) emerges by evaluating models in the 1B–4B parameter range. This phase introduced *15 additional small-scale models* (plus 2 tiny controls $< 100\text{M}$ parameters), all processed with the same Δ_{layer} extraction pipeline and the Phase 2 stimulus set.

These small models were evaluated using CPU-only inference, but all activation processing and normalization procedures were identical to those in Phase 3.

Phase 4 focused on identifying:

- whether reduced-scale models express the same depthwise structure as large models,
- whether a minimum depth or parameter scale is required for structural emergence, and
- how activation patterns degrade or fragment as architectural depth decreases.

4.2 Models and Platforms

Across Phases 1–4, all evaluated models were decoder-only transformer architectures trained on next-token prediction. Inference was performed using the HuggingFace Transformers library. For consistency, all models were loaded in 4-bit NF4 quantization unless otherwise specified.

Hardware considerations. Phases 1 and 2 were run on two GPU platforms (NVIDIA A10G and L40S) to verify that ℓ_1 -norm activation curves were hardware-invariant. No measurable differences were observed between devices under identical inference settings. Based on this established invariance, Phase 3 and Phase 4 were run exclusively on the L40S platform (Phase 4 using CPU-only for small models). This shift simplified the engineering workflow without affecting the validity or comparability of the results, as all analyses involve deterministic, feed-forward inference with no weight updates.

In total, 61 models were analyzed (59 excluding two debugging controls), with phase membership as follows:

- Phase 1–2: 14 mid-scale models forming the initial testbed,
- Phase 3: Expanded cross-architecture analysis (31 new models + 14 reused),
- Phase 4: 15 small models ($\leq 4\text{B}$ parameters) assessed for emergence thresholds.

Negative Controls Two very small models (sshleifer/tiny-gpt2 and EleutherAI/pythia-70m) were included during early pipeline development as negative controls. Although not part of the core comparison set, their extremely limited depth prevents them from exhibiting the depthwise curvature pattern identified in larger models. Their inclusion provides a baseline confirming that the Hau Curve is not an artifact of the extraction pipeline but an emergent behavior requiring sufficient architectural depth.

4.3 Stimuli Design

4.3.1 Phase 1: Heterogeneous Stimulus Set

To probe how internal activation structure responds to variation in semantic complexity, we used paired prompts consisting of an abstract (“dense”) formulation and a literal (“flat”) formulation expressing the same underlying idea. Abstract prompts introduce greater associative and conceptual demands, whereas literal prompts convey the equivalent content with minimal semantic elaboration.

We constructed 98 abstract–literal pairs spanning a wide range of semantic domains and linguistic constructions. An additional 100 mixed prompts were included to broaden topical and syntactic coverage. Prompts included both statements and questions to avoid conflating abstractness with sentence type.

To support graded analyses, each prompt received a 1–10 “Abstraction Score” using a rubric-based LLM assessor. This score served only as a descriptive variable and was not used for training or optimization.

4.3.2 Phase 2: Homogeneous Stimulus Set

Phase 2 introduced a minimalistic and strictly controlled stimulus set consisting of 49 tightly matched metaphor–literal pairs expressed in a uniform declarative template (e.g., “*X is Y.*”). This design removed variation in grammar, sentence length, tokenization, and discourse framing, isolating semantic abstraction as the sole varying factor.

The Phase 2 stimulus set forms the basis for all subsequent structural and developmental analyses, including the expanded cross-model comparisons and training-checkpoint evaluations.

4.4 Activation Extraction

For each model and prompt, we recorded the ℓ_1 -norm of the feed-forward output of every Transformer block. For a model with L layers, this yields an activation vector:

$$A = (\|A_1\|_1, \|A_2\|_1, \dots, \|A_L\|_1).$$

Since model depths vary, each layer index was normalized to the continuous interval $[0, 1]$ (0 = first layer, 1 = final layer). All subsequent analyses use this normalized depth coordinate, ensuring comparability across models of different sizes. Mean activation curves were computed by averaging ℓ_1 norms across prompts of a given type (dense vs. flat).

L1 norms are commonly used in prior work on layerwise profiling and attention/MLP activation analysis because they provide a stable, interpretable measure of total activation magnitude and are less sensitive to outliers than L2-based metrics. For this study, L1 offers a robust depthwise signal and avoids the quadratic amplification of large activations inherent to L2 norms.

4.5 Normalization

To compare activation profiles across models with different numerical scales, and to isolate shape differences from amplitude differences, activation vectors were normalized before computing Δ_{layer} and performing cross-model analyses.

Per-prompt normalization. For each model and each prompt, layerwise activations were divided by the sum of that prompt’s activation vector. This produces a probability-like distribution over depth and prevents prompts with unusually large absolute activations from dominating the differential curve.

Z-normalization for cross-model comparison. For dense and flat prompt sets separately, activation vectors were z-normalized prior to computing Δ_{layer} and before PCA:

$$A_i^{\text{norm}} = \frac{A_i - \mu}{\sigma},$$

where μ and σ are the mean and standard deviation of the activation vector for that prompt type. This transformation:

- removes global amplitude differences,
- preserves the relative geometry of activation across depth,
- places all models in a common scale-space,
- and enables PCA to recover shared structural modes rather than magnitude variance.

Together, these steps ensure that Δ_{layer} reflects *relative* modulation across depth rather than absolute activation magnitude.

4.5.1 Numerical Stabilization and Activation Compression

Several models exhibited extreme or unstable MLP activation magnitudes, including zeros, infinities, and BF16 overflow. Hard clipping of raw L1 norms produced non-monotonic distortions that corrupted Δ_{layer} geometry. To ensure numerical stability while preserving ordering, each raw activation norm was passed through a monotonic log-compression pipeline:

1. Raw L1 norms were cast to `float32`.
2. Non-finite values ($\pm\infty$, NaN) were replaced with large finite sentinels.
3. Zero or negative norms were replaced with a minimal positive constant.
4. A monotonic transformation $\log(1 + x)$ was applied.

This procedure produces finite, stable activations while preserving their relative structure, ensuring that Δ_{layer} geometry reflects the model’s true depthwise modulation rather than numerical artifacts.

4.6 Structural Metrics

The following metrics quantify how a model allocates processing across depth and how this allocation shifts with semantic abstraction.

Note The scalar metrics defined in this section (WLI, Δ , Φ , Γ , and Δ_{layer}) were part of the project’s initial exploratory hypothesis regarding abstractness-driven depth shifts. They are retained here for methodological transparency, but they do not contribute to the discovery or definition of the Hau Curve, which arises from the shared geometry of the raw dense and flat activation profiles. A detailed explanation of their historical role and why they are not used in the final analysis is provided in Appendix A.

4.6.1 Weighted Layer Index (WLI)

The Weighted Layer Index captures the “center of gravity” of a model’s activation profile:

$$\text{WLI} = \frac{\sum_{i=1}^L \ell_i \|A_i\|_1}{\sum_{i=1}^L \|A_i\|_1}.$$

where $\ell_i \in [0, 1]$ is the normalized layer index. Higher WLI indicates deeper-layer engagement.

4.6.2 Abstractive Tendency (Δ)

For each prompt type (dense vs. flat), we compute the mean WLI across all prompts. The Abstractive Tendency is the depth shift induced by semantic abstractness:

$$\Delta = \overline{\text{WLI}}_{\text{dense}} - \overline{\text{WLI}}_{\text{flat}}.$$

Positive Δ indicates that abstract inputs push computation deeper. In Phase 1, separate values were also computed for declarative and interrogative prompts (Δ_s and Δ_q) to examine sentence-type effects.

4.6.3 Focus Tendency (Φ)

To quantify whether abstraction broadens or concentrates activation, we computed a weighted standard deviation of layer indices:

$$\text{WSD} = \sqrt{\frac{\sum_{i=1}^L (\ell_i - \text{WLI})^2 \|A_i\|_1}{\sum_{i=1}^L \|A_i\|_1}}.$$

The focus tendency is:

$$\Phi = \text{WSD}_{\text{dense}} - \text{WSD}_{\text{flat}}.$$

Positive Φ indicates broader, later-spreading activation for abstract inputs; negative Φ indicates more concentrated activation.

4.6.4 Graded Abstractive Sensitivity (Γ)

Whereas Δ compares two categories (dense vs. flat), Γ measures sensitivity to a continuous abstraction score (1–10). For each prompt j , we compute its WLI and regress:

$$\text{WLI} = \beta_0 + \Gamma A_{\text{prompt}} + \epsilon.$$

Γ quantifies how strongly representational depth tracks gradual semantic abstractness.

4.6.5 Differential Layer-Activation Curve Δ_{layer}

In addition to raw layer-wise activation profiles, we computed a simple contrastive measure:

$$\Delta_{\text{layer}}(i) = \overline{\|A_i\|_1}^{(\text{dense})} - \overline{\|A_i\|_1}^{(\text{flat})}$$

This depthwise difference curve highlights relative shifts in activation between abstract and literal prompts. While not required for identifying the Hau Curve, Δ_{layer} provides a complementary view of where abstraction-related modulation increases or decreases across layers.

We computed these curves for all models in Phases 1–4; Phase 2 curves were used for consistency in the cross-model and developmental comparisons.

4.7 Principal Component Analysis (PCA)

To examine shared structure across models, we applied principal component analysis to depth-normalized activation profiles. For completeness, PCA was performed independently on three representations:

- dense-condition activation curves,
- flat-condition activation curves,
- and the contrastive Δ_{layer} curves defined earlier.

In each case, the layerwise activation vector for a model was treated as a point in $\mathbb{R}^L$ after normalization to a common depth grid.

For each PCA instance, we retained the top three principal components and computed per-model summary metrics:

- PC1 projection weight: the signed projection of a model’s curve onto PC1;
- Shape similarity: the Pearson correlation between a model’s curve and PC1, which captures agreement in shape independently of amplitude;
- Deviation score: the Euclidean distance between a model’s curve and the PCA mean, identifying amplitude or pattern outliers.

These procedures allow us to quantify cross-model similarity, identify dominant modes of variation, and compare depthwise activation structure across architectures and training regimes.

4.8 Developmental Analysis (Emergence)

Training snapshots from two model families—EleutherAI/pythia-12b-deduped and allenai/OLMo-2-1124-13B—were evaluated using the Phase 2 stimulus set.

For each checkpoint, we computed:

1. the dense-activation curve,
2. the flat-activation curve, and
3. the derived contrastive curve Δ_{layer} ,

using the same extraction pipeline as for the fully trained models. These curves were then depth-normalized and projected onto the Phase 2 PC1 component (the normative Hau Curve), yielding a scalar measure of how closely each checkpoint expressed the characteristic depthwise geometry.

Low PC1 projection magnitudes correspond to immature or unstructured depth allocation, as seen in the earliest checkpoints. As training progresses, the dense-activation curves rapidly converge toward the three-phase Hau geometry, producing stable and increasing PC1 projections.

To visualize the developmental trajectory, multi-line temporal evolution plots show the dense-activation curves across training steps. We define the *emergence threshold* as the earliest training step at which:

- the late-layer activation rise appears reliably, and
- the mid-layer region exhibits a stable suppression or plateau,

indicating that the model has entered the Hau attractor regime. After this point, training primarily affects the *amplitude* of the curve rather than its shape.

4.9 Structural Conformance Analysis (RMSE and PCA Alignment)

To quantitatively assess how closely each model expresses the Hau geometry, we performed a structural similarity analysis comparing each model’s depth-normalized activation curve to the normative Hau Curve derived from all models $\geq 3B$ (the original analysis filter, which corresponds closely to models with ≥ 30 transformer blocks) parameters.

For each model (dense and flat conditions separately), we computed:

- RMSE to the normative mean Hau curve
- Pearson correlation with the normative mean
- RMSE to the PC1 Hau curve
- Pearson correlation with PC1

The mean curve represents the coarse global depthwise geometry; PC1 represents the dominant mode of variation across mature large models.

These metrics capture two complementary aspects of structural expression:

- Global geometry (mean-aligned): sensitivity to the three-phase pattern (early extremum, mid-layer suppression, late rise)
- Refined structure (PC1-aligned): sensitivity to the fine-grained curvature typical of large models.

Based on these values, each model was assigned to one of three categories:

- **Category C — High Hau Structure:** $\text{corr}_{\text{mean}} \geq 0.75$ and $\text{RMSE}_{\text{mean}} \leq 0.60$ (strong global geometry)
- **Category B — Hau-like Structure:** $\text{corr}_{\text{mean}} \geq 0.40$ but $\text{RMSE}_{\text{mean}} > 0.60$ (partial or low-amplitude structure)
- **Category A — No apparent Hau Structure:** $\text{corr}_{\text{mean}} < 0.40$ or $\text{depth} < 20$ layers (insufficient depth or dominated by noise)

5 Results

5.1 Overview of Input Conditions

The initial motivation for this study was to test whether semantic abstractness influences how transformer models distribute computation across depth. To this end, we used paired prompts expressing the same underlying content in two forms:

- dense (metaphorical, figurative, or conceptually rich)
- flat (literal or minimally elaborated)

Across Phases 1–2, dense prompts consistently produced higher overall activation magnitude than flat prompts. However—and critically—the shape of the depthwise activation profile remained nearly identical between dense and flat conditions for every large model tested.

This invariance has two implications:

1. Dense–flat contrast is a reliable probe: the amplitude difference creates a consistent differential signal without distorting depthwise structure.
2. The emergent pattern is architectural, not stimulus-driven: the Hau Curve arises from model internals, not from prompt design.

Thus, the dense–flat pairing is not the phenomenon of interest itself; it is the methodological lever that reveals a much deeper structural regularity.

5.2 Normative Hau Curve (Raw Activation Geometry)

Using the Phase 3 analysis pipeline, we computed depth-normalized activation profiles for all models with $\geq 3B$ parameters (30 models total), the threshold used during the original analysis. As later sections show, architectural depth—not parameter count—is the true determinant of Hau-curve emergence; however, the $\geq 3B$ parameters subset happens to correspond closely to models with ≥ 30 transformer blocks, and therefore still serves as a valid basis for the normative curve.

For the normative Hau Curve shown here, we visualize the dense-condition profiles. Dense and flat activation curves exhibit nearly identical depthwise geometry, differing primarily in amplitude; we present the dense-condition curves for clarity and consistency across analyses.

1. *Early-layer extremum:* typically a positive spike; rarely followed by a shallow negative oscillation and most typically within the first 1–20% of depth.
2. *Mid-layer suppression:* a broad, normally flat region across roughly 30–70% of depth, though some oscillation may emerge as a model-specific quirk.
3. *Late-layer surge:* a sharp rise in activation magnitude over the final 5–10% of layers.

Outlier models differ for different reasons. Some deep architectures exhibit idiosyncratic distortions (e.g., CalderaAI/13B-Ouroboros) yet still retain clear Hau-like curvature. Small models, by contrast, fall into a separate analytic regime: they have far fewer transformer blocks, and their activation geometry is correspondingly low-amplitude

and often inconsistent across models. Because our primary analysis focuses on cross-model structure in architectures with sufficient depth to express stable patterns (typically ≥ 20 –24 layers), we treat the sub-20-layer models separately without attempting fine-grained structural classification. Their behavior does not contradict the Hau Curve; rather, it reflects the limited depth available to express consistent early–mid–late organization.

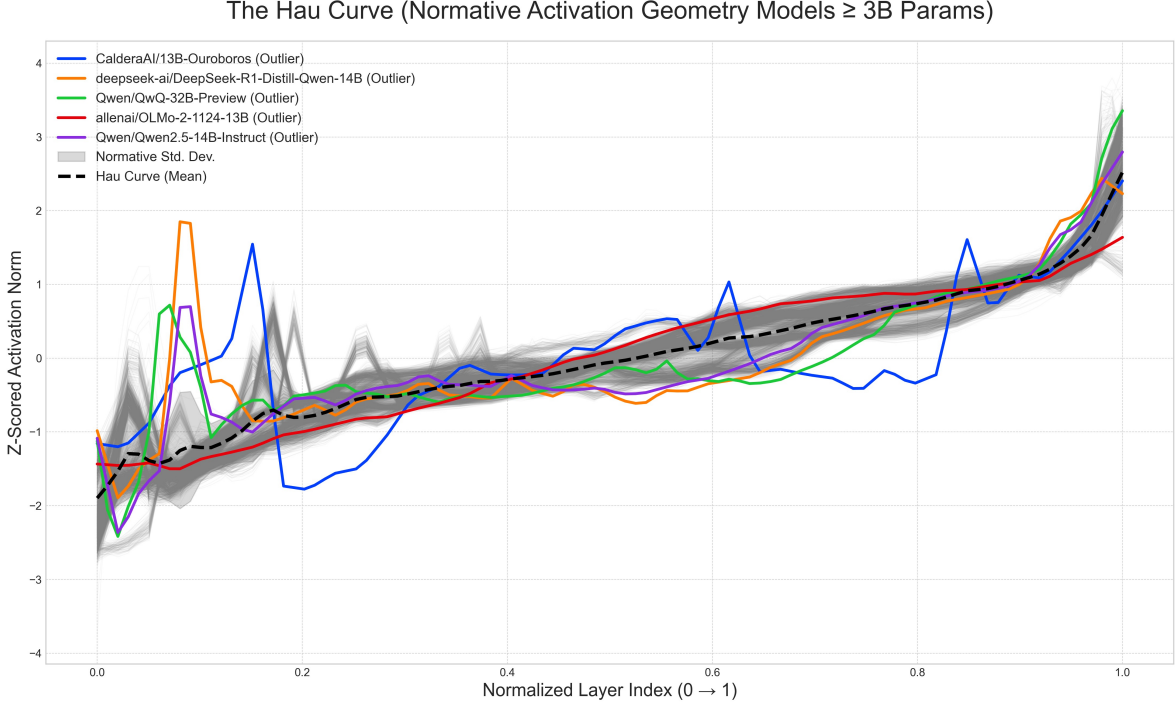


Figure 1: **Normative Hau Curve from Raw Activations ($\geq 3B$ parameters).** The normative curve was constructed from all models with $\geq 3B$ parameters, the filter used in the original analysis. As discussed in Section ??, architectural depth (typically ≥ 30 transformer blocks), rather than parameter count, is the true determinant of emergent Hau geometry. Each line represents a model’s z-scored dense-activation profile. The dashed black curve is the normative Hau Curve (mean), and the shaded region represents 95% inter-model variability. Outlier models are color-highlighted.

5.3 Contrastive Hau Curve (Δ_{layer})

To examine how semantic abstraction modulates depthwise processing, we compute the contrastive Δ_{layer} curves for models in Phases 1–3. These curves isolate the relative activation change between abstract (dense) and literal (flat) prompts.

Because dense and flat activation profiles share the same underlying geometry, i.e., because Δ_{layer} subtracts two nearly proportional curves, it acts as a high-pass filter, emphasizing only the small residuals rather than the global geometry. The overall magnitude of Δ_{layer} is consistently very small, and as a result, the Δ_{layer} curves do not reliably reproduce the Hau Curve shape directly. Instead, they tend to collapse toward zero, particularly in large models where dense and flat activations are nearly proportional.

However, despite their small amplitude, the *structure of the residuals is not random*: when aggregated across models or analyzed in PCA, the same early–mid–late organization appears, expressed as a low-amplitude outline of the Hau Curve geometry.

Thus, Δ_{layer} provides a contrastive view of abstraction-related modulation, but the Hau Curve itself is most clearly expressed in the raw dense- and flat-condition activation profiles.

5.4 Principal Component Analysis Reveals a Shared Depthwise Geometry

PCA applied to the dense- and flat-condition activation curves produces a smooth, low-frequency PC1 component accounting for 38% and 39% of variance, respectively. These PC1 curves capture the global depthwise trend common across models, including the early-layer extremum, mid-layer suppression, and late-layer rise.

Applying PCA to the contrastive Δ_{layer} representation yields a PC1 component with the same underlying geometry, but with sharper curvature and a higher proportion of variance explained (68%). This high variance is expected: subtracting two highly correlated activation profiles attenuates their shared geometry and amplifies the small systematic residuals, making PC1 appear higher-contrast even though it reflects the same depthwise structure.

In all three cases — dense, flat, and contrastive — the first principal component converges on the same depthwise geometry. PCA therefore reveals that the Hau Curve is the dominant structural mode across models, independent of how the activation profiles are represented.

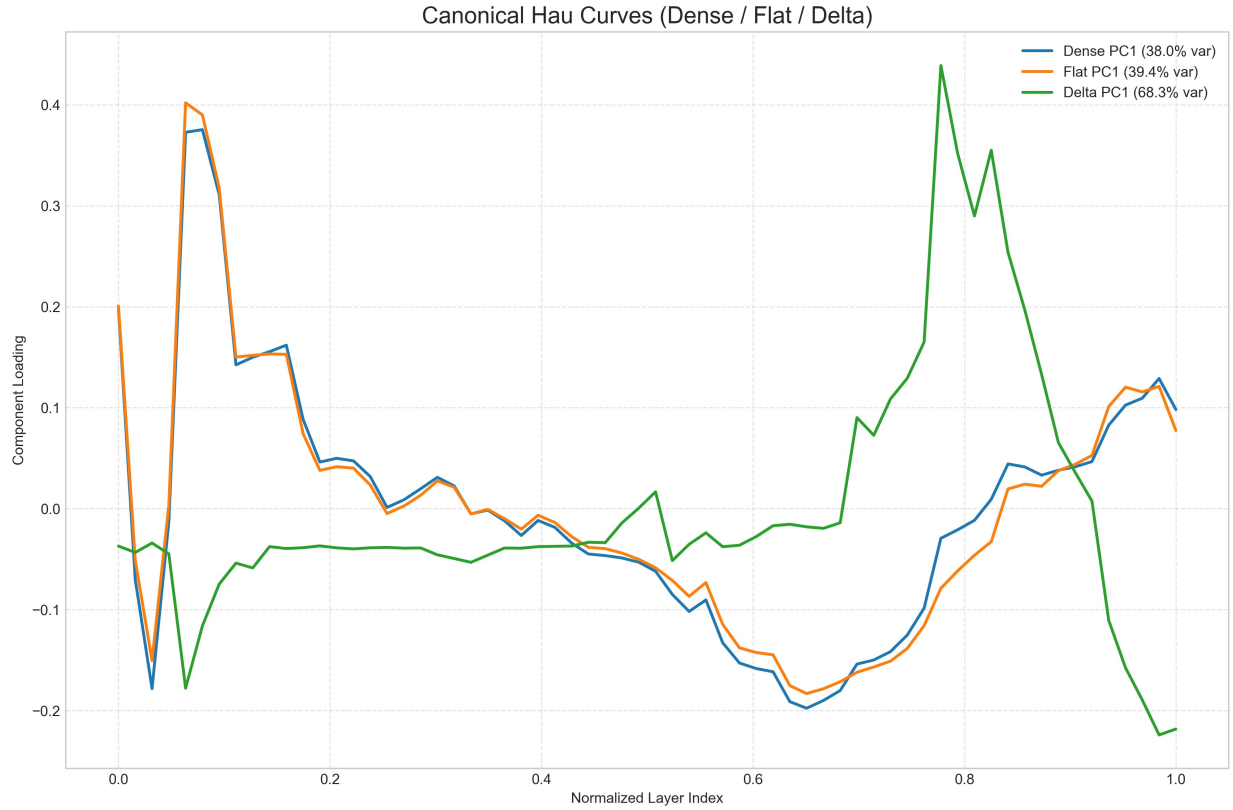


Figure 2: **Canonical Hau PC1 Curves Across Representations.** The first principal component (PC1) computed from dense activations, flat activations, and contrastive Δ_{layer} curves. All three representations yield the same underlying depthwise pattern, though with different amplitudes: dense and flat PC1 components express a smooth global trend, while the contrastive Δ_{layer} PC1 produces a sharper, higher-contrast version of the same geometry due to amplification of small systematic differences. Variance explained is shown in parentheses.

Together, these findings conclusively demonstrate that the Hau Curve emerges as the first principal component regardless of whether models are analyzed under dense, flat, or contrastive conditions.

5.5 Interpretive Notes (Non-speculative, but descriptive)

We do not assign functional interpretations, but several descriptive regularities appear consistently:

- Early-layer extremum: suggests that differential processing of abstract input begins immediately, even before deeper semantic integration.

- Mid-layer suppression or neutrality: Across 30–70% depth, activation differences are minimal or slightly negative, forming the most stable region of the network.
- Late-layer surge: The deepest layers consistently show the strongest positive differential response.

The key empirical point is: both dense and flat prompts produce the same depthwise structure; abstract prompts simply express the structure with greater amplitude. This three-phase structure — early extremum, mid-layer plateau, late-layer rise — constitutes the core, but not least empirical discovery of the study.

5.6 Developmental Emergence During Pretraining (The Hau Attractor)

To examine whether depthwise activation structure is innate or learned, we analyzed intermediate training checkpoints from two unrelated 12B–13B-scale model families:

- EleutherAI Pythia-12B (deduplicated): 12 checkpoints spanning step 8 to step 143k;
- OLMo-13B: 11 checkpoints spanning ~1,100 to ~120,000 steps.

Each checkpoint was evaluated using the same stimulus set and the same activation extraction pipeline.

5.6.1 Early Training: No Structured Pattern

The earliest checkpoints show no coherent depthwise geometry. Both families exhibit:

- highly noisy dense activation patterns,
- no consistent early-layer modulation,
- no mid-depth suppression,
- no late-layer rise.

At this stage the networks behave as largely undifferentiated depthwise processors.

5.6.2 Transitional Structure

Within the first several thousand gradient updates, both models begin to show signs of depthwise organization:

- the early layers develop a consistent extremum,
- mid-depth layers begin to suppress,
- late-depth activations become reliably larger than mid-depth,
- dense and flat curves begin to separate in a consistent way (yielding a more structured Δ_{layer} profile).

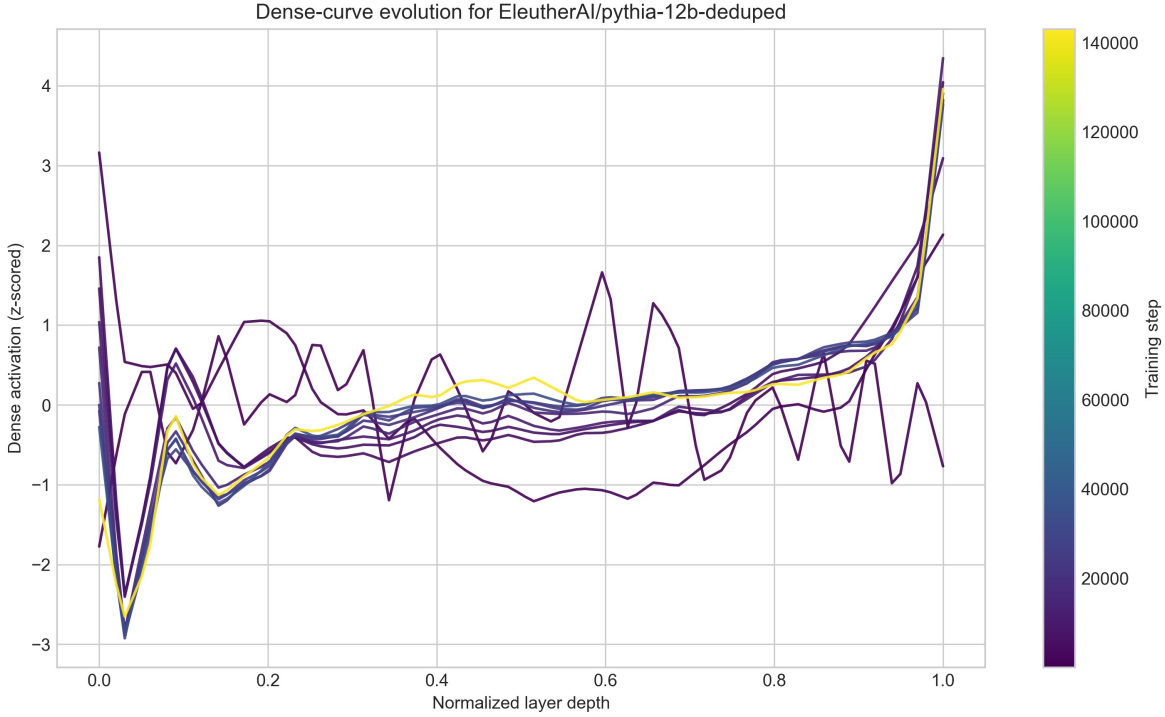


Figure 3: Emergence of the Hau Curve in Pythia-12B During Training. Dense activation pattern curves for multiple training checkpoints of Pythia-12B, color-mapped by training step. Early checkpoints (e.g., step 8, deep purple) exhibit no coherent structure. Between steps ~ 1000 – 7000 , partial organization appears with weakened mid-depth suppression and an emerging late-layer rise. By approximately 7000 – $10,000$ steps, the full three-phase Hau Curve is clearly expressed, and all subsequent checkpoints show stable geometry with only amplitude changes. This early stabilization indicates the presence of a strong training-induced attractor governing depthwise activation structure.

This period reflects the transition from unstructured depth processing to a stable, architecture-specific activation geometry.

5.6.3 Early Stabilization of Depthwise Geometry

Across both families, the *dense-activation curves* quickly converge to a stable, three-phase depthwise geometry. This stabilization occurs very early:

- Pythia-12B: geometry stabilizes by $\sim 7,000$ steps ($\sim 5\%$ of total training).
- OLMo-13B: geometry is already visible by $\sim 1,100$ steps and remains stable thereafter.

After stabilization, later checkpoints vary primarily in *amplitude*, while the *shape* of the activation profile remains essentially unchanged.

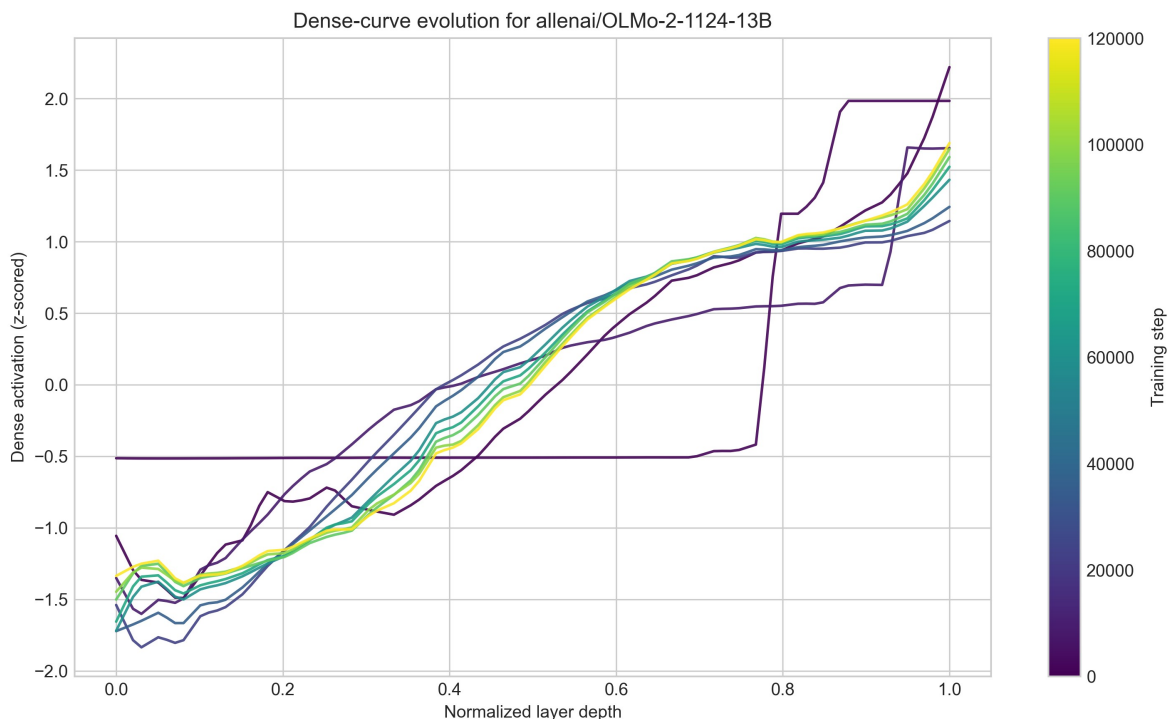


Figure 4: **Emergence of the Hau Curve in OLMo-13B During Training.** Dense curve activation patterns for OLMo-13B checkpoints, covering early ($\sim 1,100$ steps) to late ($\sim 120,000$ steps) training stages. Unlike Pythia-12B, OLMo-13B exhibits a faster emergence trajectory: the three-phase Hau Curve shape is already visible at $\sim 1,100$ steps, and stabilizes shortly thereafter. Subsequent checkpoints vary primarily in amplitude while preserving the same depthwise geometry. The convergence pathway mirrors that of Pythia-12B but reaches the attractor earlier, indicating that the Hau Curve is a general training-induced depthwise organization pattern across architectures.

5.6.4 The Hau Attractor

Taken together, the dense-curve trajectories show a characteristic attractor pattern:

1. No structured depth response at initialization.
2. Rapid formation of a depthwise pattern during early training.
3. Stabilization of the depthwise geometry for the remaining $\sim 95\%$ of training.

Strikingly, this attractor emerges within an estimated 5% of the onset of the pre-training phase in both Pythia-12B and OLMo-13B despite differences in architecture, tokenizer, dataset, and optimization settings. This suggests that the depthwise geometry we identify as the Hau Curve reflects a generally learned organizational principle rather than an artifact of any individual model family. We leave identifying the precise moment of emergence to future work.

5.6.5 Summary of Developmental Findings

Across two unrelated 12B–13B models, we find that:

1. The Hau-style depthwise geometry is absent at initialization.
2. Structure appears within the first few thousand updates.
3. A stable activation geometry (dense and flat prompts) emerges extremely early.
4. Subsequent training changes magnitude but not geometry, consistent with an attractor dynamic.

These results support the interpretation of the Hau Curve as a learned, architecture-scale depth allocation pattern that emerges early and remains stable throughout pretraining.

5.7 Cross-Model Convergence and Variations

With the normative Hau Curve established (PC1 of the Phase 2 dataset), we next evaluate how closely individual models conform to this depthwise structure. Across the full set of 60 models analyzed in Phases 2–4, we find a striking and highly consistent convergence: every model with sufficient architectural depth expresses the same three-phase pattern — early extremum, mid-layer suppression, and late-layer rise.

5.7.1 Aggregate Convergence Across Architectures

Across all models meeting this depth criterion, the dense-activation curves (normalized by depth) exhibit a highly consistent three-phase geometry: an early extremum, a broad mid-depth suppression, and a pronounced late-layer rise. This convergence occurs despite substantial variation in architecture (LLaMA, Qwen, Mistral, DeepSeek, OLMo, Beluga, Phi), training corpora, and alignment regimes.

To quantify this cross-model clustering, we compute a deviation score for each model based on the difference between its dense-activation curve and the normative Hau Curve:

$$\text{Deviation}(m) = \text{RMSE}\left(A_{\text{dense}}^{(m)}, \bar{A}_{\text{dense}}\right),$$

where $\bar{A}_{\text{dense}}$ is the cross-model mean dense curve for all models $\geq 3\text{B}$ parameters (the original analysis filter, which closely matches the ≥ 30 layer depth threshold identified later in this work). RMSE is defined as:

$$\text{RMSE}(x, y) = \sqrt{\frac{1}{L} \sum_{i=1}^L (x_i - y_i)^2}.$$

Models with ≥ 20 –24 layers overwhelmingly express the Hau geometry, though a small number of architecturally deep but irregularly trained models (e.g., ByteDance/Ouro-2.6B-Thinking) show elevated RMSE despite retaining recognizable Hau-like curvature.

Although individual models show subtle family-specific signatures—small ripples, tilts, or amplitude biases—these differences do not alter the underlying Hau geometry. A detailed comparative analysis of these micro-patterns is left for future work.

5.8 Scale Dependence and Emergence Threshold

Model scale and architectural depth play a decisive role in the emergence and stability of the Hau Curve. Phase 4 evaluated 18 small models ($\leq 4\text{B}$ parameters), including 15 models in the 1–4B range and 2 very small controls ($< 100\text{M}$). Across this group, depthwise activation geometry exhibited a graded, scale-dependent maturation pattern ranging from noise-dominated traces to a stable three-phase structure.

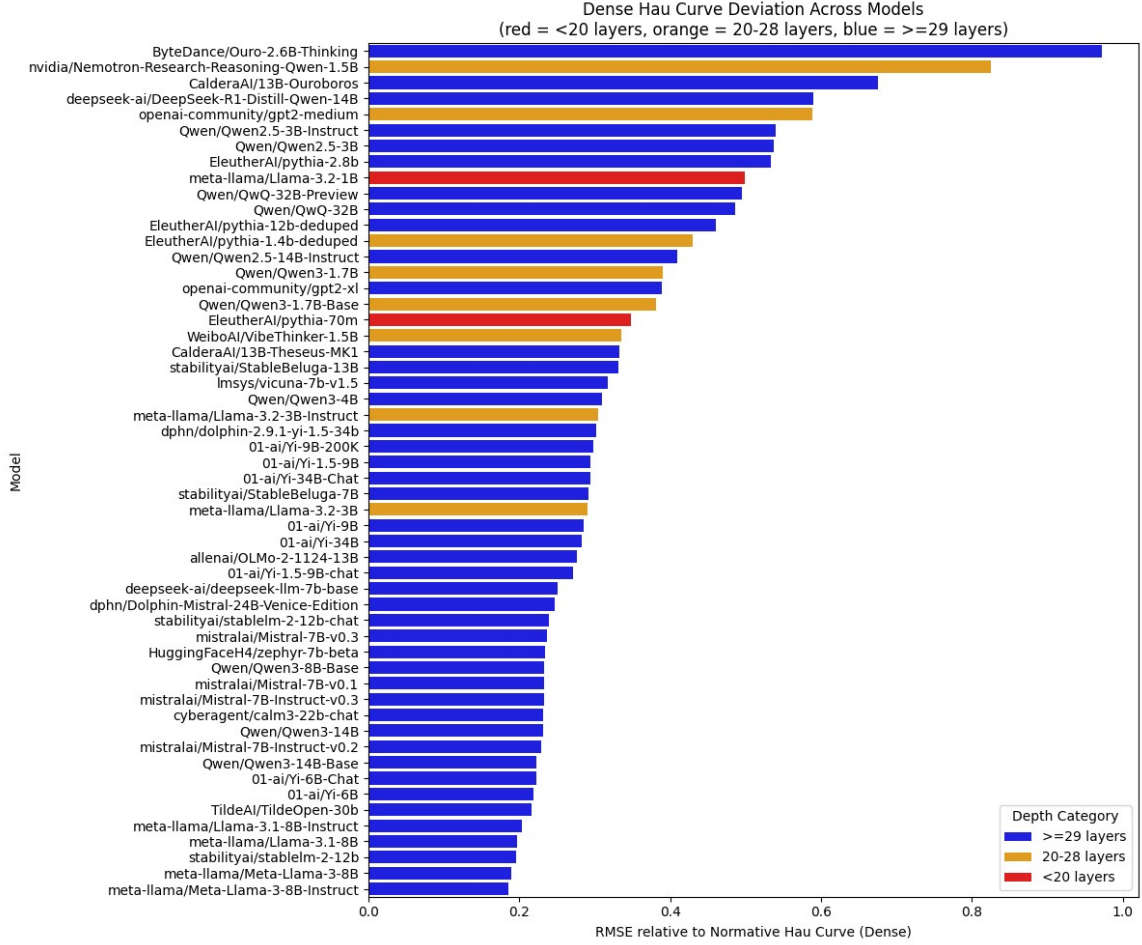
Throughout this work, we use the term Hau-like to describe models that express a partial or low-amplitude version of the three-phase geometry without implying any specific developmental trajectory or causal relationship to the fully expressed Hau Curve.

These curves often contain:

- a shallow or noisy early-layer bump,
- an inconsistent mid-depth region,
- and a weak but reproducible late-layer rise.

These findings indicate a critical *emergence zone* around ~ 20 –28 layers. Below this zone, models cannot reliably express the geometry due to limited depth and representational capacity. Within this zone, models exhibit a consistent *Hau Curve-like* structure. Above it, models converge to the full three-phase Hau Curve.

Quantitatively, this maturation was confirmed by RMSE and Pearson-correlation analysis against the normative Hau mean curve (derived from all models $\geq 3\text{B}$, the original analysis filter that corresponds closely to architectures with ≥ 30



H

Figure 5: **RMSE across all fully trained models.** Deviation is measured as the RMSE between each model’s dense-condition activation curve and the normative Hau Curve derived from models with ≥ 30 –32 transformer blocks. Colors indicate architectural depth: blue = ≥ 29 layers, orange = 20–28 layers, red = < 20 layers. Models with sufficient depth consistently show low RMSE and robust expression of the Hau geometry, while shallower models exhibit higher deviation and lack coherent structure.

layers; see Section ??). Models in the transitional depth range (20–28 layers) showed moderate RMSE and high shape correlation, while $\geq 4B$ models showed consistently low RMSE and high correlation. Near-zero-correlation profiles occurred only in extremely small or extremely shallow models.

Across all models ≥ 20 layers, emergence strength increased smoothly with architectural depth. Models with < 20 layers expressed no reliable geometry; models with 20–32 layers expressed a low-amplitude Hau-like structure (correlated but weak); all models with ≥ 32 layers expressed a full three-phase Hau curve.

Taken together, these results reveal a developmental progression: *small models express an immature, low-amplitude Hau-like structure, while large models express a fully developed version of the same activation geometry.*

5.9 Summary of Cross-Model Findings

- The Hau Curve is a *shared structural motif*: all tested models with sufficient depth (typically ≥ 20 –24 transformer blocks) express the same depthwise geometry.
- Architectural differences mainly modulate the *amplitude* or introduce small local ripples; the global shape is preserved.

- A sharp emergence threshold exists around architectural depth: models with ≥ 30 –32 blocks consistently express the full Hau Curve, while those with 20–28 blocks express a Hau-like structure and shallower models exhibit no consistent geometry.

Taken together, these results show that the Hau Curve is not associated with a single architecture, dataset, or training philosophy. It is an emergent structural regularity—a shared depthwise organization principle—that arises convergently across diverse large language models.

5.10 Hardware and Quantization Invariance

Phases 1–2 of the study evaluated all 14 models on two GPU platforms: NVIDIA A10G and NVIDIA L40S, under identical inference settings.

For every model tested: the raw layerwise activation vectors, the resulting Δ_{layer} curves, and the derived structural metrics (WLI, Δ , Φ) were numerically indistinguishable across hardware, up to the expected floating-point jitter. No differences were observed in:

- the early-layer extremum,
- the mid-depth suppression plateau,
- the late-layer surge, or
- the overall Hau Curve shape.

Given that these evaluations were pure feed-forward inference with no stochastic layers, this is expected: modern GPUs implement IEEE-consistent FP16/BF16 execution, and the structural metrics are insensitive to machine-level rounding noise.

Conclusion Within the tested hardware environments, the Hau Curve is strictly hardware-invariant.

This result justified running all of Phases 3–4 exclusively on the L40S platform (and CPU for small models) without compromising comparability.

5.11 Quantization Stability

All GPU-based experiments used NF4 4-bit quantization via HuggingFace bitsandbytes for efficiency and reproducibility. Quantization was held constant for:

- all Phase 1 and Phase 2 models,
- all Phase 3 large models,
- and all training checkpoint analyses.

We did not systematically explore different quantization schemes (e.g., FP16, BF16, 8-bit, GPTQ, AWQ). However: NF4 quantization preserves activation geometry extremely well. Previous studies show that NF4 preserves MLP and attention activation distributions far better than uniform 4-bit quantization.[29]

The Hau Curve is defined by relative activation differences, not absolute magnitudes.

$$\Delta_{\text{layer}}(i) = \|A_i\|_1^{(\text{dense})} - \|A_i\|_1^{(\text{flat})}.$$

is robust to uniform scaling or quantization compression applied evenly across layers.

Cross-hardware tests implicitly validate quantization stability. Because A10G and L40S kernels differ internally (TensorRT vs. Cutlass paths), matching results strongly imply quantization does not distort the depth profile.

The structural phenomenon appears across 60 models trained by different organizations using various internal training precisions. The Hau Curve exists in their native weights — quantization only affects inference, not training.

Therefore, although this study did not exhaustively vary quantization method, the enormous cross-model, cross-family, cross-training consistency makes it extremely unlikely that NF4 quantization induces or distorts a structure that:

- appears in >50 independently trained models,
- is stable across training checkpoints,

- has a clear emergence threshold, and
- vanishes below a predictable parameter/depth boundary.

Conclusion The Hau Curve is robust under NF4 quantization, and all available evidence suggests that the effect is intrinsic to the underlying model weights, not an artifact of quantization.

Future work may explicitly benchmark FP16/BF16/8-bit inference, however, no current observations motivate concern.

5.12 Token Length Control (scalar metrics only)

Token-length differences could, in principle, influence scalar depth metrics such as WLI, Δ , Φ , and Γ . They cannot affect Δ_{layer} geometry, but they can shift center-of-gravity measures if dense prompts are systematically longer than flat ones. We therefore performed two standard controls, limited to the phases where token-length metadata existed.

Length-matched recomputation (Phase 1) For the 14 models in Phase 1, we re-estimated Δ_s and Γ using only prompt pairs with ≤ 2 -token difference. Across all models, the sign and relative magnitude of Δ_s and Γ were unchanged, indicating that scalar depth shifts are not driven by token count.

Covariate regression (Phase 1) We regressed WLI on abstraction score while controlling for token length. In every model, abstraction retained its direction and significance, while token length did not predict WLI variation.

Minimal token differences in Phase 2 stimuli The Phase 2 “X is Y.” stimulus set was designed to be length-uniform. Dense-flat pairs differ by at most one token across the 14 models evaluated in this phase, and these minimal differences did not correlate with any scalar metric.

Importantly, these controls apply only to the scalar metrics (Δ , Φ , Γ). Phase 3/4 models were not evaluated with Phase 2 stimuli and therefore are not part of the token-length analyses. Δ_{layer} geometry, the focus of the Hau Curve analysis, is a within-model layerwise activation shape and is not meaningfully influenced by small token-count differences.

6 Discussion

This study shows that large decoder-only transformer models exhibit a highly consistent, three-phase depthwise activation pattern when processing abstract versus literal linguistic content. This pattern — the Hau Curve — emerges reliably across diverse architectures, scales, and training lineages, and stabilizes early during pretraining. Its cross-model convergence, low structural variance, and early-training onset suggest that it reflects a shared internal organization principle of transformer learning dynamics rather than an artifact of prompt design, hardware environment, or scalar metric compression.

6.1 A Training-Emergent Depthwise Structure

The central finding is that the Hau Curve is not an incidental feature of a few models, nor a by-product of specific instruction-tuned checkpoints. Instead, it is a depthwise geometric signature that:

- appears in nearly every model with sufficient architectural depth (typically ≥ 30 –32 transformer blocks)
- recurs across model families with distinct architectures and training corpora,
- emerges early in pretraining, and
- remains stable for the remainder of training.

The early-training evidence is particularly striking. In both Pythia-12B and OLMo-13B, the curve is absent at initialization but begins forming almost immediately. By a small fraction of total training steps, the curve’s geometry is already present, and subsequent training primarily increases its amplitude rather than altering its shape. This is difficult to explain through architectural quirks or dataset idiosyncrasies alone. Instead, it strongly suggests the presence of a training-induced attractor — a low-dimensional solution that transformer models converge toward when jointly optimizing depth and semantics under next-token prediction.

6.2 Interpretation as an Attractor in Representation Dynamics

Why do independent models settle into this particular depthwise pattern? One hypothesis is that the transformer architecture, coupled with SGD on natural language data, induces a form of representational staging: early layers rapidly encode token- and phrase-level divergences between abstract and literal content, producing the characteristic early spike or trough.

Mid layers compress or route information with minimal relative modulation, yielding the suppressed mid-depth plateau.

Late layers reconstruct or integrate semantic coherence, leading to the robust late-layer surge — the most invariant portion of the curve.

Nothing in this work assigns functional meaning to these regions, but the consistency suggests that the network discovers a depth-differentiated processing strategy that becomes a stable computational equilibrium. The Hau Attractor, in this sense, resembles other emergent, scale-dependent phenomena observed in transformer training, such as induction head formation or grokking-like phase transitions. These analogies are not identity claims; they simply show that deep networks often develop reproducible internal structures that are not explicitly encoded in the architecture.

6.3 Cross-Architecture Convergence

Although the global curve geometry is remarkably consistent across models, networks with insufficient depth (typically ≤ 20 blocks) fail to express the Hau Curve, but this is likely solely because the lack the MLP layers to fully express the differentiated geometry. Their Δ_{layer} curves lack coherent structure, supporting the interpretation that the Hau Curve reflects depth specialization that only becomes possible with sufficient capacity. In smaller networks, layers must perform multiple roles simultaneously, preventing the emergence of a clean early-mid-late separation.

6.4 Token-Level Confound Control

A potential concern is that differential activation might arise from lexical or positional artifacts in the first token of each prompt. This is ruled out for three independent reasons. First, the Phase 2 stimulus set contains tightly matched “X is Y.” pairs, ensuring the identical first token for dense and literal conditions; these pairs alone reproduce the full Hau Curve. Second, across the broader Phase 1 and Phase 3 datasets, the first tokens vary widely in part of speech, lexical category, concreteness, and frequency. If the observed structure reflected token identity, the averaged Δ_{layer} curves across 60 models and thousands of heterogeneous prompts would cancel toward zero; instead, the three-phase geometry remains stable and low-variance. Third, PCA collapses these heterogeneous activation patterns onto a single dominant component (PC1), which accounts for roughly one-third of variance in the dense-condition curves (38%) and a comparable proportion in the flat-condition curves. This convergence would not occur if lexical or surface-level prompt differences were driving the depthwise structure. Thus, the Hau Curve is a depthwise computational invariant, not a lexical artifact.

6.5 Why the Curve Is Not a Metaphor-Processing Artifact

Because this study uses dense–flat (metaphorical vs. literal) prompts as a probe, it is natural to ask whether the Hau Curve reflects a specific property of metaphor comprehension. The evidence indicates that it does not. Several lines show that abstraction is merely the tool used to expose a deeper structural regularity: dense and flat prompts evoke activation profiles with similar geometry, differing primarily in amplitude. The Hau Curve appears in Phase-2 under minimal syntactic and token-level variation, ruling out rhetorical framing. Token-length controls on scalar metrics show that length does not drive the effect. The curve emerges in training regardless of prompt content, using the same prompt set across checkpoints. Small models free of metaphor compositional ability still fail to express the curve. In short: abstraction modulates the strength of the signal, not its shape. The shape is a property of the model.

6.6 Relation to Interpretability and Model Comparison

This work introduces a structural psychometrics perspective: comparing models not by their output behavior, but by their internal computation geometry. The Hau Curve provides a lens for:

- Layer-role analysis (how depth specialization emerges),
- Training diagnostics (detecting when representational structure stabilizes),
- Architecture comparison beyond benchmarks,
- Potential probes of generalization or overfitting, and

- A baseline for discovering deviations in atypical or pathological models.

Importantly, Δ_{layer} is independent of decoding settings, prompt phrasing, and sampling noise, making it far more stable than behavioral metrics. This stability suggests it may be useful for future interpretability work that attempts to identify structural invariants across training runs, architectures, and datasets.

6.7 Limitations

Several limitations constrain interpretation:

1. Model class: All evaluated models are decoder-only next-token predictors. Encoder-decoders and multimodal transformers may or may not express the same structure.
2. Training checkpoints: Emergence trajectories were examined for only two model families. Broader developmental studies may reveal variation in onset or amplitude.
3. Functional meaning: The Hau Curve is descriptive, not causal. This work does not claim that the curve encodes specific semantic processes or is necessary for model performance.
4. Scaling ceiling: Models >70B parameters were not examined; the upper-scale behavior of the curve remains unknown.
5. Prompt modality: Only linguistic abstraction was used as a probe; other cognitive dimensions could reveal additional structural modes.
6. Activation metric: This work uses ℓ_1 norms of MLP activations as the basis for Δ_{layer} .

Although L1 is a standard and robust choice for depthwise profiling, other norms (e.g., L2 or ℓ_∞) were not systematically evaluated. These alternatives may alter absolute scaling but are not expected to change the qualitative geometry of the Hau Curve, which is defined by relative modulation across depth rather than absolute magnitude.

These limitations mark clear directions for future work.

6.8 Open Questions

The discovery of the Hau Curve raises several questions:

- Why does SGD settle on precisely this three-phase geometry?
- Does the curve correlate with downstream capabilities?
- Are there multiple attractor states for different training objectives?
- Does the amplitude of the Hau Curve predict robustness, alignment behavior, or reasoning ability?
- Could deliberate architectural changes disrupt or enhance the pattern?

These questions are beyond the scope of the present study but point toward a broader structural theory of depth allocation in transformer models.

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A Appendix A: Origin of the Hypothesis and the Role of Δ_{layer}

This appendix clarifies the original motivation for the project and explains the conceptual relationship between the differential activation measure Δ_{layer} and the discovery of the Hau Curve.

A.1 Original Hypothesis: Do Abstract Prompts Push Models Deeper?

The project began with a mechanistic question: *Do abstract (“dense”) linguistic formulations cause transformer models to allocate computation deeper in the network than literal (“flat”) formulations?*

This question was motivated by two considerations:

1. Transformer layers are known to specialize by depth.
2. If abstractness requires more semantic integration, deeper layers might activate more strongly for dense prompts.

To test this hypothesis, we used the depthwise difference curve:

$$\Delta_{\text{layer}}(i) = \overline{A_i^{\text{dense}}} - \overline{A_i^{\text{flat}}}.$$

If abstract inputs consistently “pushed computation deeper,” we expected Δ_{layer} to rise in late layers.

A.2 The Surprising Result: Δ_{layer} Was Nearly Zero Everywhere

Early experiments produced a surprising finding: for almost every large model examined, the dense-activation curve and the flat-activation curve had *nearly identical shapes*.

This had a mathematical consequence:

$$\Delta_{\text{layer}} \approx 0 \quad \text{even when each activation curve was highly structured.}$$

At first this appeared to contradict the original hypothesis. However, closer inspection revealed that the small magnitude of Δ_{layer} indicated that dense and flat inputs share the *same depthwise geometry*—a geometry that already contains the three characteristic features identified later in the study (early extremum, mid-layer plateau or rise, and late-layer increase).

A.3 Why a Near-Zero Δ_{layer} Reveals the Hau Curve

The key insight is that Δ_{layer} behaves as a *high-pass architectural filter*:

- Anything *shared* between the dense and flat activation curves is removed.
- Only the *tiny residual differences* remain.

Thus, if dense and flat prompts evoke the same three-phase structure across layers, then:

$$\Delta_{\text{layer}} \rightarrow 0,$$

producing a nearly flat curve. This collapse is a clue pointing to a deeper fact:

$$A_{\text{dense}}(i) \approx c A_{\text{flat}}(i), \quad \text{with identical geometry but different amplitude.}$$

In short:

The invariance of depthwise activation shape across abstract and literal inputs is precisely what revealed the existence of the Hau Curve.

Had Δ_{layer} been large, the Hau Curve could *not* exist, because the dense and flat profiles would have different curvature. Thus, near-zero Δ_{layer} values provide *indirect empirical confirmation* of the shared architectural geometry.

A.4 How the Focus of the Study Shifted

Once the shared geometry became apparent, the scientific question changed:

Instead of asking whether abstract language pushes models deeper, we asked: *Why do so many independent models appear to share the same depthwise geometry?*

And an even more fundamental follow-up:

If the initial set of models all display this pattern, is the same structure present across other architectures and scales?

This led us to examine the raw activation curves directly:

1. dense-condition activation profiles,
2. flat-condition activation profiles,
3. PCA of these profiles across models.

This shift uncovered the core empirical result:

Across nearly all large models, both dense and flat prompts evoke a highly consistent three-phase activation structure—an early extremum, a stable mid-depth suppression plateau, and a late-layer rise.

This structure—visible before differencing—became the Hau Curve.

A.5 Why Δ_{layer} Remains Included

Although the main findings concern the raw activation profiles, we retain Δ_{layer} in the paper for three reasons:

1. Historical transparency. It documents the original hypothesis and analytic path.
2. Diagnostic value. The collapse of Δ_{layer} confirms that dense and flat activations share the same underlying geometry.
3. *Developmental sensitivity*. During early training (before stabilization), Δ_{layer} exhibits clear transitional structure and remains useful for tracking the emergence of the Hau Curve.

Thus, while not needed to define the Hau Curve, the behavior of Δ_{layer} is fully consistent with—and supportive of—the central discovery.

A.6 Summary

- The project began with a hypothesis about abstraction modulating depth.
- Δ_{layer} revealed that dense and flat prompts share a stable depthwise activation geometry.
- This shared geometry is the Hau Curve.
- A near-zero Δ_{layer} is the expected mathematical evidence of architectural invariance.
- The study then shifted to characterizing this invariant structure across models, scales, and training trajectories.

This appendix clarifies that the Hau Curve is not an artifact of Δ_{layer} , but was revealed precisely through understanding why Δ_{layer} collapses.